

An energy company is considering an investment in wind-power production. To determine whether this investment is economically feasible, the company seeks to estimate the expected revenue of a wind-power producer operating in a competitive spot market.

Consider a day-ahead spot market with five electricity producers. Three producers generate electricity from conventional sources, such as coal, whereas the remaining two rely on solar and wind power, respectively. The market is cleared once each day for 24 hourly trading intervals of the following day. For each interval, every producer submits a bid that specifies the electricity quantity it is willing to supply (MWh), together with the minimum price it is willing to accept ($\$/\text{MWh}$). The market operator then applies a merit-order clearing mechanism to determine both the market-clearing price and the dispatch schedule, subject to the requirement that total electricity demand be satisfied. If a producer's bid is accepted, the producer is obligated to supply the nominated quantity and is remunerated at the market-clearing price. If the bid is not accepted, the producer receives no revenue for that interval.

The energy company intends to develop a model to estimate the expected revenue of the wind-power producer on any future day by determining the hourly bid quantities and prices under uncertainty. On this basis, the model can subsequently be extended to examine outcomes across a range of short-term and long-term scenarios, thereby supporting an assessment of the investment's economic viability.

Please formalise this problem as an agent-based model, and specify the key components, including time scales, agents, state variables, control variables, processes, objectives, constants, uncertain parameters and stochastic variables. At this stage, the model should be limited to simulating single-day market operations.

Show less

Agent-Based Model: Wind-Power Producer in a Day-Ahead Spot Market

1. Overview

The model simulates a single day of operations in a day-ahead electricity spot market. Five producer agents submit bids for each of 24 hourly intervals; a market operator agent clears each interval via a merit-order mechanism; and the wind-power producer's daily revenue is computed as the aggregate of accepted-bid revenues across all intervals.

2. Time Scales

Scale	Definition
Planning horizon	A single calendar day d
Trading interval	Each hour $t \in \mathcal{T} = \{1, 2, \dots, 24\}$
Bid submission	Once per day, prior to interval $t = 1$, covering all $T = 24$ intervals simultaneously
Market clearing	Once per interval t , after all bids are received

3. Agents

3.1 Producer Agents

Let $\mathcal{N} = \mathcal{N}^C \cup \mathcal{N}^S \cup \mathcal{N}^W$ denote the full set of producers, where:

- $\mathcal{N}^C = \{c_1, c_2, c_3\}$ — conventional (coal) producers

- $\mathcal{N}^S = \{s\}$ — solar producer
- $\mathcal{N}^W = \{w\}$ — wind producer *(the decision-making agent of interest)*

3.2 Market Operator Agent

A passive clearing agent that receives all bids, applies the merit-order mechanism for each interval t , and returns clearing prices and dispatch decisions. It has no strategic behaviour; it is a deterministic process.

4. State Variables

State variables describe the condition of each agent and the market at any given interval t .

4.1 Wind Producer State

Variable	Symbol	Units	Description
Realised wind output	$\tilde{q}_t^W$	MWh	Actual generation available in interval t , revealed after bidding
Accumulated daily revenue	$R_d^W = \sum_{\tau=1}^{t-1} r_\tau^W$	\$	Revenue earned up to but not including interval t
Dispatch status	$\delta_t^W \in \{0, 1\}$	—	1 if bid accepted, 0 otherwise

4.2 Conventional Producer States

For each $i \in \mathcal{N}^C$:

Variable	Symbol	Units	Description
Available capacity	K_t^i	MWh	Capacity online in interval t (subject to outage)
Dispatch status	$\delta_t^i \in \{0, 1\}$	—	1 if bid accepted

4.3 Solar Producer State

Variable	Symbol	Units	Description
Realised solar output	$\tilde{q}_t^S$	MWh	Actual generation available, correlated with irradiance forecast
Dispatch status	$\delta_t^S \in \{0, 1\}$	—	1 if bid accepted

4.4 Market State

Variable	Symbol	Units	Description
Market-clearing price	λ_t	\$/MWh	Price resulting from merit-order clearing in interval t
Total demand	D_t	MWh	Aggregate electricity demand to be satisfied in interval t

5. Control Variables

Control variables are the decision variables set by each producer agent prior to market clearing.

For each producer $i \in \mathcal{N}$ and each interval $t \in \mathcal{T}$, producer i submits a bid:

$$b_t^i = (q_t^i, p_t^i)$$

Variable	Symbol	Units	Description
Bid quantity	q_t^i	MWh	Quantity offered for supply in interval t
Bid price	p_t^i	\$/MWh	Minimum acceptable price in interval t

For the wind producer specifically, the bid is the primary modelling focus:

$$b_t^W = (q_t^W, p_t^W), \quad t \in \mathcal{T}$$

Since the wind producer has near-zero marginal cost, p_t^W is typically set at or near zero (a price-taking strategy), making q_t^W the principal control variable. At the bidding stage, q_t^W must be determined under uncertainty about $\tilde{q}_t^W$.

6. Processes

6.1 Bid Formation (per producer, before $t = 1$)

Each producer i determines its bid schedule $\{b_t^i\}_{t \in \mathcal{T}}$ based on its type:

- **Conventional producers:** bid at marginal fuel cost mc^i with full available capacity K_t^i
- **Solar producer:** bids expected generation $\mathbb{E}[\tilde{q}_t^S]$ at a near-zero price
- **Wind producer:** bids a quantity q_t^W based on a wind forecast, at price $p_t^W \geq 0$

6.2 Merit-Order Market Clearing (per interval t)

1. Sort all accepted bids in ascending order of bid price to form the **supply stack**
2. Identify the **marginal bid**: the lowest-priced bid whose cumulative quantity meets or first exceeds D_t
3. Set the ****market-clearing price****: $\lambda_t = p_t^{i^*}$, where i^* is the marginal producer
4. **Dispatch**: all producers with $p_t^i \leq \lambda_t$ are dispatched ($\delta_t^i = 1$); others are not

Formally:

$$\lambda_t = \min \left\{ p_t^i : \sum_{j: p_t^j \leq p_t^i} q_t^j \geq D_t \right\}$$

$$\delta_t^i = \mathbf{1} [p_t^i \leq \lambda_t]$$

6.3 Revenue Realisation (per interval t)

The wind producer's revenue in interval t is:

$$r_t^W = \delta_t^W \cdot \lambda_t \cdot \min(q_t^W, \tilde{q}_t^W)$$

The $\min(\cdot)$ term accounts for potential under-delivery if realised output falls short of the bid quantity. Daily revenue is:

$$R_d^W = \sum_{t=1}^{24} r_t^W$$

6.4 Stochastic Processes

- **Wind generation:** $\tilde{q}_t^W$ is drawn from a stochastic process (see Section 8)
- **Solar generation:** $\tilde{q}_t^S$ is drawn from a correlated diurnal process
- **Demand:** D_t follows a diurnal profile with random perturbations

7. Objectives

7.1 Wind Producer Objective

The wind producer seeks to maximise expected daily revenue:

$$\max_{\{q_t^W, p_t^W\}_{t \in \mathcal{T}}} \mathbb{E} [R_d^W] = \mathbb{E} \left[\sum_{t=1}^{24} \delta_t^W \cdot \lambda_t \cdot \min(q_t^W, \tilde{q}_t^W) \right]$$

subject to:

$$0 \leq q_t^W \leq \bar{Q}^W \quad \forall t \in \mathcal{T}$$

$$p_t^W \geq 0 \quad \forall t \in \mathcal{T}$$

where $\bar{Q}^W$ is the installed wind capacity.

7.2 Conventional Producer Objectives

Each $i \in \mathcal{N}^C$ maximises profit by bidding at marginal cost (competitive price-taking behaviour):

$$p_t^i = mc^i, \quad q_t^i = K_t^i$$

8. Constants and Parameters

8.1 Structural Constants

Symbol	Description	Units
$T = 24$	Number of trading intervals per day	—
$\ \mathcal{N}\ = 5$	Total number of producers	—
$\bar{Q}^W$	Installed wind capacity	MWh
$\bar{Q}^S$	Installed solar capacity	MWh
$\bar{K}^i$	Nameplate capacity of conventional producer i	MWh

8.2 Techno-Economic Parameters

Symbol	Description	Units
mc^i	Marginal cost of conventional producer i	\$/MWh
$mc^W \approx 0$	Marginal cost of wind producer	\$/MWh
$mc^S \approx 0$	Marginal cost of solar producer	\$/MWh
$\bar{D}_t$	Baseline (expected) demand in interval t	MWh

9. Uncertain Parameters and Stochastic Variables

These are quantities whose values are unknown at bid-submission time and must be modelled probabilistically.

Symbol	Type	Description	Suggested Distribution
$\tilde{q}_t^W$	Stochastic	Wind generation in interval t	$\text{Beta}(\alpha_t, \beta_t) \times \bar{Q}^W$, capturing diurnal and seasonal shape; or a truncated normal on $[0, \bar{Q}^W]$
$\tilde{q}_t^S$	Stochastic	Solar generation in interval t	Zero for night hours; $\text{Beta}(\alpha_t^S, \beta_t^S) \times \bar{Q}^S$ for daylight hours
$\tilde{D}_t$	Stochastic	Total demand in interval t	$\mathcal{N}(\bar{D}_t, \sigma_t^D)$, truncated to $[D^{\min}, D^{\max}]$
$\tilde{K}_t^i$	Uncertain	Available capacity of conventional producer i	Bernoulli outage model: $\tilde{K}_t^i = \bar{K}^i$ w.p. $(1 - \phi^i)$, else 0
$\tilde{p}_t^{-W}$	Emergent	Clearing price conditional on wind not being marginal	Derived from realisations of $\tilde{D}_t, \tilde{q}_t^S, \tilde{K}_t^i$

Key dependency structure:

$\tilde{q}_t^W \not\perp \tilde{q}_{t+1}^W$ (temporally correlated via wind speed process)

$\tilde{q}_t^W \not\perp \tilde{q}_t^S$ (negatively correlated: cloud cover affects both)

$\tilde{D}_t \not\perp \tilde{q}_t^S$ (positively correlated: solar irradiance drives cooling load)

10. Summary Diagram

```

DAY-AHEAD BID SUBMISSION (once, before  $t = 1$ )
|
| Each agent  $i$  forms bid schedule  $\{(q_{t^i}, p_{t^i})\}$  for  $t = 1..24$ 
| Wind producer uses wind forecast  $E[\tilde{q}_{t^W}]$  under uncertainty
|
└─> FOR EACH INTERVAL  $t = 1..24$ :
    |
    |─ Stochastic variables realised:  $\tilde{q}_{t^W}, \tilde{q}_{t^S}, \tilde{D}_t, K_{t^i}$ 
    |
    |─ Market Operator: Merit-order clearing
    |   Sort bids by price → find marginal bid → set  $\lambda_t$ 
    |   Dispatch:  $\delta_{t^i} = 1$  if  $p_{t^i} \leq \lambda_t$ 
    |
    |─ Revenue:  $r_{t^W} = \delta_{t^W} \cdot \lambda_t \cdot \min(q_{t^W}, \tilde{q}_{t^W})$ 
    |
DAILY OUTCOME:  $R_{d^W} = \sum_{t=1}^{24} r_{t^W}$ 

```

11. Modelling Assumptions (Single-Day Scope)

1. **No inter-temporal constraints:** ramp rates, start-up costs, and storage are excluded at this stage
2. **Perfect competition:** all producers are price-takers; no strategic manipulation of the supply stack
3. **Inelastic demand:** D_t is exogenous and must be met exactly; no demand response
4. **No transmission constraints:** the market is a single copper-plate node
5. **No balancing market:** imbalance penalties for under-delivery are simplified via the $\min(q_t^W, \tilde{q}_t^W)$ term
6. **Bid commitment:** bids submitted once and cannot be revised within the day

These assumptions define a tractable baseline. Each can be relaxed in subsequent extensions — for example, by introducing a balancing market, strategic bidding behaviour, or multi-day scenarios for the longer-term investment assessment.

You can assume that the market demand follows a normal distribution with constant mean and variance. Please refine your specification accordingly and present the revised section.

9. Uncertain Parameters and Stochastic Variables (*Revised*)

These are quantities whose values are unknown at bid-submission time and must be modelled probabilistically.

Symbol	Type	Description	Distribution
$\tilde{q}_t^W$	Stochastic	Wind generation in interval t	$\text{Beta}(\alpha_t, \beta_t) \times \bar{Q}^W$, capturing diurnal shape on $[0, \bar{Q}^W]$
$\tilde{q}_t^S$	Stochastic	Solar generation in interval t	Zero for night hours; $\text{Beta}(\alpha_t^S, \beta_t^S) \times \bar{Q}^S$ for daylight hours

Symbol	Type	Description	Distribution
$\tilde{D}_t$	Stochastic	Total demand in interval t	$\tilde{D}_t \sim \mathcal{N}(\mu^D, \sigma^{D^2})$, i.i.d. across all $t \in \mathcal{T}$
$\tilde{K}_t^i$	Uncertain	Available capacity of conventional producer i	Bernoulli outage model: $\tilde{K}_t^i = \bar{K}^i$ w.p. $(1 - \phi^i)$, else 0
$\tilde{\lambda}_t$	Emergent	Market-clearing price in interval t	Derived from realisations of all stochastic variables via the merit-order process

The demand specification warrants three remarks.

Distributional form. The assumption $\tilde{D}_t \sim \mathcal{N}(\mu^D, \sigma^{D^2})$ i.i.d. for all t implies that demand carries no diurnal pattern, no serial autocorrelation, and no cross-interval dependence. This is a strong simplification relative to observed load profiles, but it is adopted here as a tractable baseline that isolates the effect of supply-side uncertainty on the wind producer's revenue.

Support constraint. Since electricity demand is strictly positive, the normal distribution should be treated as approximately valid only when the probability of drawing a non-positive realisation is negligible, i.e. when $\mu^D \gg \sigma^D$. Formally, one may impose a positivity constraint by conditioning on $\tilde{D}_t > 0$, giving a truncated normal; however, provided μ^D / σ^D is large (e.g. a coefficient of variation below 0.10–0.15), the truncation correction is immaterial and the untruncated form is retained for analytical convenience.

New constants. The two distributional parameters are added to the set of structural constants in Section 8.2:

Symbol	Description	Units
μ^D	Mean electricity demand, constant across all intervals	MWh
σ^{D^2}	Variance of electricity demand, constant across all intervals	MWh ²

Dependency structure. The i.i.d. demand assumption eliminates the inter-variable correlation $\tilde{D}_t \not\perp\!\!\!\perp \tilde{q}_t^S$ noted in the original specification. Under this revision, demand is independent of all other stochastic variables, and the remaining dependencies are:

$$\tilde{D}_t \perp \tilde{q}_t^W, \tilde{q}_t^S, \tilde{K}_t^i \quad \forall t \in \mathcal{T}$$

$$\tilde{q}_t^W \not\perp\!\!\!\perp \tilde{q}_{t+1}^W \quad (\text{temporally correlated via wind speed process})$$

$$\tilde{q}_t^W \not\perp\!\!\!\perp \tilde{q}_t^S \quad (\text{negatively correlated via shared cloud-cover exposure})$$

You can also make the following assumptions regarding the behavioural logic of the producers: (1) conventional producers have stable bid quantities, with bid prices following normal distributions with uncertain means and variances; (2) the solar-power producer's bid quantity follows $\max(0, a + b \cos(\frac{2\pi t}{24}))$, with bid prices following a normal distribution with uncertain mean and variance; and (3) actual wind-power production is stochastic, following a normal distribution with constant mean and variance. Please refine your specification accordingly and present the revised section.

Revised Sections 5, 6.1, and 9

The three behavioural assumptions jointly affect the control variables, the bid-formation process, and the stochastic parameter table. All three sections are revised below.

5. Control Variables (Revised)

For each producer $i \in \mathcal{N}$ and each interval $t \in \mathcal{T}$, the submitted bid remains $b_t^i = (q_t^i, p_t^i)$. The determination of each component now differs by producer type as follows.

5.1 Conventional Producers ($i \in \mathcal{N}^C$)

Bid quantities are stable and equal to each producer's installed capacity in every interval:

$$q_t^i = \bar{K}^i, \quad \forall t \in \mathcal{T}$$

Bid prices are drawn from producer-specific normal distributions:

$$p_t^i \sim \mathcal{N}(\mu_p^i, (\sigma_p^i)^2), \quad \forall t \in \mathcal{T}$$

where μ_p^i and σ_p^i are themselves uncertain parameters (see Section 9). Since negative prices are theoretically permissible in some markets but would be inconsistent with cost-recovery behaviour, it is assumed that $\mu_p^i \gg \sigma_p^i$, so that the probability of drawing $p_t^i \leq 0$ is negligible.

5.2 Solar Producer ($i = s$)

The bid quantity follows a truncated cosine profile that approximates the diurnal irradiance cycle:

$$q_t^S = \max\left(0, a + b \cos\left(\frac{2\pi t}{24}\right)\right), \quad \forall t \in \mathcal{T}$$

where $a \in \mathbb{R}$ is a baseline offset and $b > 0$ is the amplitude. The $\max(0, \cdot)$ operator enforces non-negative generation; intervals for which $a + b \cos(2\pi t/24) \leq 0$ correspond to night hours with zero solar output. The bid price follows:

$$p_t^S \sim \mathcal{N}(\mu_p^S, (\sigma_p^S)^2), \quad \forall t \in \mathcal{T}$$

with μ_p^S and σ_p^S uncertain (see Section 9).

5.3 Wind Producer ($i = w$)

The wind producer's bid (q_t^W, p_t^W) remains the primary set of control variables. The bid quantity q_t^W must be determined at submission time under uncertainty about actual production $\tilde{q}_t^W$, which is realised only after bids are submitted. The bid price $p_t^W \geq 0$ is a decision variable, typically set near zero given near-zero marginal cost.

$$0 \leq q_t^W \leq \bar{Q}^W, \quad p_t^W \geq 0, \quad \forall t \in \mathcal{T}$$

6.1 Bid Formation Process (Revised)

Producer	Bid Quantity q_t^i	Bid Price p_t^i
Conventional $i \in \mathcal{N}^C$	$\bar{K}^i$ (fixed constant)	Drawn from $\mathcal{N}(\mu_p^i, (\sigma_p^i)^2)$
Solar s	$\max(0, a + b \cos(2\pi t/24))$ (deterministic function of t)	Drawn from $\mathcal{N}(\mu_p^S, (\sigma_p^S)^2)$
Wind w	q_t^W (optimised decision, see Section 7.1)	p_t^W (optimised decision, see Section 7.1)

The removal of the Bernoulli outage model for conventional producers (previously in Section 9) means that supply-side quantity uncertainty now originates solely from the wind producer's realised output $\tilde{q}_t^W$. Price uncertainty on the supply side enters through the stochastic bid prices of conventional and solar producers.

9. Uncertain Parameters and Stochastic Variables (Revised)

9.1 Stochastic Variables

Symbol	Description	Distribution
$\tilde{q}_t^W$	Actual wind-power production in interval t	$\mathcal{N}(\mu^W, (\sigma^W)^2)$, i.i.d. across t
$\tilde{D}_t$	Total market demand in interval t	$\mathcal{N}(\mu^D, (\sigma^D)^2)$, i.i.d. across t
$\tilde{p}_t^i$	Bid price of conventional producer i in interval t	$\mathcal{N}(\mu_p^i, (\sigma_p^i)^2)$, i.i.d. across t
$\tilde{p}_t^S$	Bid price of solar producer in interval t	$\mathcal{N}(\mu_p^S, (\sigma_p^S)^2)$, i.i.d. across t
$\tilde{\lambda}_t$	Market-clearing price in interval t	Emergent; derived from merit-order clearing given all realisations

9.2 Uncertain Parameters

The means and variances governing the bid-price distributions are themselves uncertain — reflecting, for instance, incomplete knowledge of competitors' cost structures — and must be estimated or calibrated from historical data:

Symbol	Description	Units
μ_p^i	Mean bid price of conventional producer i	\$/MWh
$(\sigma_p^i)^2$	Variance of bid price of conventional producer i	(\$/MWh) ²
μ_p^S	Mean bid price of solar producer	\$/MWh
$(\sigma_p^S)^2$	Variance of bid price of solar producer	(\$/MWh) ²

9.3 Additional Constants

The following parameters are added to Section 8.2 to support the revised behavioural rules:

Symbol	Description	Units
a	Baseline offset in solar bid-quantity profile	MWh
b	Amplitude of solar bid-quantity profile	MWh
μ^W	Mean actual wind-power production	MWh
$(\sigma^W)^2$	Variance of actual wind-power production	MWh ²

9.4 Support Constraints and Approximation Remarks

Both $\tilde{q}_t^W$ and $\tilde{D}_t$ are physically bounded below by zero. As with demand (Section 9, previous revision), the normal approximation is valid provided the coefficient of variation is small: $\sigma^W/\mu^W \ll 1$ and $\sigma^D/\mu^D \ll 1$. If this condition does not hold, a truncated normal on $[0, \bar{Q}^W]$ and $[0, \infty)$ respectively should be substituted.

9.5 Revised Dependency Structure

The i.i.d. assumption applied uniformly across all stochastic variables eliminates all inter-variable and inter-temporal dependencies from the original specification. The dependency structure simplifies to full mutual independence:

$$\tilde{q}_t^W \perp \tilde{D}_t \perp \tilde{p}_t^i \perp \tilde{p}_t^S, \quad \forall i \in \mathcal{N}^C, \forall t \in \mathcal{T}$$

$$\tilde{q}_t^W \perp \tilde{q}_{t'}^W, \quad \forall t \neq t'$$

This represents a deliberate simplification: in practice, wind output exhibits strong temporal autocorrelation, and competitor bid prices are likely correlated across intervals. These dependencies can be reintroduced in subsequent model extensions without altering the single-day structural framework.

When multiple bids at the clearing price collectively exceed the residual demand, producers are dispatched in priority order: solar $\succ$ wind $\succ$ conv1 $\succ$ conv2 $\succ$ conv3, with the last accepted producer potentially dispatched only partially to exactly meet demand. Please refine your specification accordingly and present the revised section.

6.2 Merit-Order Market Clearing (Revised)

6.2.1 Notation

Let $\pi = (s, w, c_1, c_2, c_3)$ denote the fixed priority ordering used to resolve ties at the clearing price. For each interval t , define the ordered set of all producers by ascending bid price, with ties broken according to π .

6.2.2 Clearing Procedure

For each interval $t \in \mathcal{T}$, the market operator executes the following steps.

Step 1 — Rank bids. Sort all bids in ascending order of bid price. Ties in bid price are broken by the priority ordering π , so that $s \succ w \succ c_1 \succ c_2 \succ c_3$ among producers submitting identical prices.

Step 2 — Identify the marginal unit. Traverse the sorted supply stack and accumulate quantity until demand is met. Let $\sigma(k)$ denote the producer ranked k -th in the stack, and define the cumulative supplied quantity after the first k producers as:

$$Q_k^{\text{cum}} = \sum_{j=1}^k q_t^{\sigma(j)}$$

The marginal producer is the producer ranked k^* , where:

$$k^* = \min \left\{ k : Q_k^{\text{cum}} \geq \tilde{D}_t \right\}$$

Step 3 — Set the clearing price. The market-clearing price is the bid price of the marginal producer:

$$\lambda_t = p_t^{\sigma(k^*)}$$

Step 4 — Determine dispatch quantities. All producers ranked strictly above the marginal unit are dispatched in full; the marginal producer is dispatched only for the residual demand; producers ranked below are not dispatched:

$$\tilde{q}_t^{\sigma(k), \text{disp}} = \begin{cases} q_t^{\sigma(k)} & \text{if } k < k^* \\ \tilde{D}_t - Q_{k^*-1}^{\text{cum}} & \text{if } k = k^* \\ 0 & \text{if } k > k^* \end{cases}$$

where $Q_0^{\text{cum}} \equiv 0$ by convention. Note that $0 < \tilde{q}_t^{\sigma(k^*), \text{disp}} \leq q_t^{\sigma(k^*)}$ whenever the marginal producer is only partially dispatched.

Step 5 — Determine dispatch indicators. The binary dispatch indicator for producer i is:

$$\delta_t^i = \mathbf{1} \left[\tilde{q}_t^{i, \text{disp}} > 0 \right]$$

so that $\delta_t^i = 1$ for all $k \leq k^*$ and $\delta_t^i = 0$ for all $k > k^*$.

6.2.3 Revised Revenue Realisation

The wind producer's revenue in interval t must now account for three distinct cases: not dispatched, partially dispatched as the marginal unit, or fully dispatched as an infra-marginal unit. In all cases, revenue cannot exceed the value of actual production:

$$r_t^W = \lambda_t \cdot \min \left(\tilde{q}_t^{W, \text{disp}}, \tilde{q}_t^W \right)$$

where $\tilde{q}_t^{W,\text{disp}} \in \{0, \tilde{D}_t - Q_{k^*-1}^{\text{cum}}, q_t^W\}$ depending on whether the wind producer is, respectively, not dispatched, the marginal unit, or infra-marginal. The $\min(\cdot)$ operator retains the imbalance protection introduced previously: if actual wind output $\tilde{q}_t^W$ falls short of the dispatched quantity, revenue is limited to the value of what was physically delivered.

6.2.4 Role of the Priority Ordering

The priority rule $s \succ w \succ c_1 \succ c_2 \succ c_3$ is consequential only when two or more producers submit bids at exactly λ_t . In this event, renewable producers (s and w) are granted preferential dispatch over conventional producers, consistent with renewable energy priority dispatch conventions observed in many real-world electricity markets. The ordering among conventional producers ($c_1 \succ c_2 \succ c_3$) serves as a deterministic tie-breaker to ensure a unique market outcome in every simulated interval.

In practice, because bid prices $\tilde{p}_t^i$ are drawn from continuous normal distributions, exact ties occur with probability zero. The priority rule therefore has negligible effect on expected revenue under the current stochastic specification, but is retained to guarantee well-posedness of the clearing mechanism for all possible realisations, including any deterministic or discretised scenarios considered in future model extensions.

Given that actual wind-power production is stochastic, it may fall short of the quantity committed to dispatch one day in advance. If the wind-power producer under-delivers relative to its dispatched quantity, the shortfall incurs a constant penalty cost (\$/MWh). Please incorporate this penalty mechanism into your specification.

Then, please provide a complete model specification that specifies all key components, including the time scales, agents, state variables, control variables, processes, objectives, constants, uncertain parameters and stochastic variables.

Complete Agent-Based Model Specification: Wind-Power Producer in a Day-Ahead Spot Market

1. Overview

The model simulates a single day of operations in a day-ahead electricity spot market. Five producer agents submit bids for each of 24 hourly intervals before the day begins. A market operator agent clears each interval via a merit-order mechanism with a fixed priority rule for tie-breaking. The wind producer's actual output is stochastic and may fall short of its dispatched quantity, in which case a per-unit penalty is applied. The wind producer's daily net revenue is the sum of hourly revenues less any hourly penalty costs.

2. Time Scales

Scale	Definition
Planning horizon	A single calendar day d
Trading interval	Each hour $t \in \mathcal{T} = \{1, 2, \dots, 24\}$
Bid submission	Once per day, prior to interval $t = 1$, covering all $T = 24$ intervals simultaneously
Market clearing	Once per interval t , after all bids are received
Delivery and settlement	Once per interval t , after physical delivery; penalties assessed at settlement

3. Agents

3.1 Producer Agents

Let $\mathcal{N} = \mathcal{N}^C \cup \{s\} \cup \{w\}$ denote the full set of five producers:

- $\mathcal{N}^C = \{c_1, c_2, c_3\}$ — conventional (coal) producers
- $\{s\}$ — solar producer
- $\{w\}$ — wind producer *(the decision-making agent of primary interest)*

Each producer is a self-interested agent that submits a bid schedule once per day to maximise its own expected payoff. Conventional and solar producers follow fixed behavioural rules; the wind producer's bidding strategy is the central object of optimisation.

3.2 Market Operator Agent

A passive, deterministic clearing agent that receives all bids, applies the merit-order mechanism with priority tie-breaking for each interval t , and returns market-clearing prices and dispatch quantities. The market operator enforces the penalty rule at settlement. It has no strategic behaviour or objective of its own.

4. State Variables

State variables describe the condition of each agent and the market at the time of, or following, clearing in interval t .

4.1 Wind Producer State

Variable	Symbol	Units	Description
Realised wind output	$\tilde{q}_t^W$	MWh	Actual generation available in interval t , revealed after bidding
Dispatched quantity	$\tilde{q}_t^{W,\text{disp}}$	MWh	Quantity the wind producer is obligated to deliver, set by clearing
Delivery shortfall	$\tilde{e}_t^W = \max(0, \tilde{q}_t^{W,\text{disp}} - \tilde{q}_t^W)$	MWh	Under-delivery relative to dispatch obligation in interval t
Interval net revenue	r_t^W	\$	Revenue less penalty in interval t
Accumulated daily net revenue	$R_d^W = \sum_{\tau=1}^{t-1} r_\tau^W$	\$	Net revenue earned up to but not including interval t

4.2 Conventional Producer States

For each $i \in \mathcal{N}^C$:

Variable	Symbol	Units	Description
Dispatched quantity	$\tilde{q}_t^{i,\text{disp}}$	MWh	Quantity dispatched in interval t
Dispatch indicator	$\delta_t^i \in \{0, 1\}$	—	1 if producer i is dispatched in interval t

4.3 Solar Producer State

Variable	Symbol	Units	Description
Dispatched quantity	$\tilde{q}_t^{S,\text{disp}}$	MWh	Quantity dispatched in interval t
Dispatch indicator	$\delta_t^S \in \{0, 1\}$	—	1 if solar producer is dispatched in interval t

4.4 Market State

Variable	Symbol	Units	Description
Market-clearing price	λ_t	\$/MWh	Price resulting from merit-order clearing in interval t
Realised demand	$\tilde{D}_t$	MWh	Aggregate electricity demand to be satisfied in interval t

5. Control Variables

Control variables are the decision variables set by each producer prior to market clearing. All bids are submitted once, before $t = 1$, and cover all 24 intervals.

For producer $i \in \mathcal{N}$ and interval $t \in \mathcal{T}$, the bid is:

$$b_t^i = (q_t^i, p_t^i)$$

Variable	Symbol	Units	Description
Bid quantity	q_t^i	MWh	Quantity offered for supply in interval t
Bid price	p_t^i	\$/MWh	Minimum acceptable price in interval t

For the wind producer, the full bid schedule $\{(q_t^W, p_t^W)\}_{t \in \mathcal{T}}$ is the optimisation object. Because the marginal cost of wind is near zero, p_t^W is typically set at or near zero, making q_t^W the principal control variable. This quantity must be committed one day in advance, before $\tilde{q}_t^W$ is realised, exposing the producer to penalty risk.

6. Processes

6.1 Bid Formation

Each producer submits its bid schedule prior to $t = 1$ according to the following rules.

Producer	Bid Quantity q_t^i	Bid Price p_t^i
Conventional $i \in \mathcal{N}^C$	$\bar{K}^i$ (fixed, full capacity)	$\tilde{p}_t^i \sim \mathcal{N}(\mu_p^i, (\sigma_p^i)^2)$
Solar s	$\max(0, a + b \cos(2\pi t/24))$	$\tilde{p}_t^S \sim \mathcal{N}(\mu_p^S, (\sigma_p^S)^2)$
Wind w	$q_t^W \in [0, \bar{Q}^W]$ (optimised)	$p_t^W \geq 0$ (optimised)

Bid prices for conventional and solar producers are drawn i.i.d. across intervals, with means and variances that are uncertain parameters estimated from historical data.

6.2 Merit-Order Market Clearing

For each interval $t \in \mathcal{T}$, the market operator executes the following procedure.

Step 1 — Rank bids. Sort all five bids in ascending order of bid price. Ties are broken by the fixed priority ordering:

$$\pi : s \succ w \succ c_1 \succ c_2 \succ c_3$$

Let $\sigma(k)$ denote the producer at rank $k \in \{1, \dots, 5\}$ after sorting.

Step 2 — Identify the marginal unit. Traverse the stack and accumulate quantity. Define cumulative supply after the first k ranked producers as:

$$Q_k^{\text{cum}} = \sum_{j=1}^k q_t^{\sigma(j)}, \quad Q_0^{\text{cum}} \equiv 0$$

The marginal rank is:

$$k^* = \min \left\{ k : Q_k^{\text{cum}} \geq \tilde{D}_t \right\}$$

Step 3 — Set the clearing price:

$$\lambda_t = p_t^{\sigma(k^*)}$$

Step 4 — Determine dispatch quantities:

$$\tilde{q}_t^{\sigma(k), \text{disp}} = \begin{cases} q_t^{\sigma(k)} & \text{if } k < k^* \\ \tilde{D}_t - Q_{k^*-1}^{\text{cum}} & \text{if } k = k^* \\ 0 & \text{if } k > k^* \end{cases}$$

The marginal producer may thus be partially dispatched, with $0 < \tilde{q}_t^{\sigma(k^*), \text{disp}} \leq q_t^{\sigma(k^*)}$.

Step 5 — Set dispatch indicators:

$$\delta_t^i = \mathbf{1} \left[\tilde{q}_t^{i, \text{disp}} > 0 \right]$$

Note on tie probability. Since bid prices are drawn from continuous normal distributions, exact ties occur with probability zero under the baseline stochastic specification. The priority rule π nonetheless ensures a unique, well-defined outcome for all realisations, including any deterministic scenarios considered in future extensions.

6.3 Physical Delivery and Shortfall Calculation

After market clearing, each dispatched producer delivers electricity. For the wind producer, actual output $\tilde{q}_t^W$ is realised from its stochastic process. The delivery shortfall is:

$$\tilde{e}_t^W = \max\left(0, \tilde{q}_t^{W, \text{disp}} - \tilde{q}_t^W\right)$$

If $\tilde{q}_t^W \geq \tilde{q}_t^{W, \text{disp}}$, the producer meets its obligation in full and $\tilde{e}_t^W = 0$. Any surplus generation beyond the dispatched quantity is not remunerated.

6.4 Settlement and Penalty Assessment

The wind producer's net revenue in interval t is:

$$r_t^W = \lambda_t \cdot \left(\tilde{q}_t^{W, \text{disp}} - \tilde{e}_t^W \right) - \rho \cdot \tilde{e}_t^W$$

The first term is revenue from physically delivered electricity, valued at the clearing price. The second term is the penalty for under-delivery, assessed at the constant penalty rate ρ . Substituting the shortfall definition, this can be written equivalently as:

$$r_t^W = \lambda_t \cdot \min\left(\tilde{q}_t^{W, \text{disp}}, \tilde{q}_t^W\right) - \rho \cdot \max\left(0, \tilde{q}_t^{W, \text{disp}} - \tilde{q}_t^W\right)$$

The daily net revenue is:

$$R_d^W = \sum_{t=1}^{24} r_t^W$$

7. Objectives

7.1 Wind Producer Objective

The wind producer chooses its bid schedule to maximise expected daily net revenue:

$$\max_{\{q_t^W, p_t^W\}_{t \in \mathcal{T}}} \mathbb{E}[R_d^W] = \mathbb{E} \left[\sum_{t=1}^{24} \left(\lambda_t \cdot \min\left(\tilde{q}_t^{W, \text{disp}}, \tilde{q}_t^W\right) - \rho \cdot \max\left(0, \tilde{q}_t^{W, \text{disp}} - \tilde{q}_t^W\right) \right) \right]$$

subject to:

$$0 \leq q_t^W \leq \bar{Q}^W, \quad p_t^W \geq 0, \quad \forall t \in \mathcal{T}$$

The penalty term creates a fundamental trade-off in the choice of q_t^W : bidding a higher quantity increases the probability of dispatch and raises potential revenue, but also increases expected penalty costs if output falls short. The optimal bid quantity therefore balances marginal expected revenue against marginal expected penalty exposure.

7.2 Conventional and Solar Producer Objectives

Conventional and solar producers follow fixed behavioural rules (Sections 6.1) and have no explicit optimisation problem within this model. Their bidding behaviour is treated as exogenous from the perspective of the wind producer.

8. Constants and Fixed Parameters

8.1 Structural Constants

Symbol	Description	Units
$T = 24$	Number of trading intervals per day	—
$\ \mathcal{N}\ = 5$	Total number of producers	—
$\pi = (s, w, c_1, c_2, c_3)$	Fixed priority ordering for tie-breaking	—

8.2 Techno-Economic Parameters

Symbol	Description	Units
$\bar{Q}^W$	Installed wind capacity	MWh
$\bar{Q}^S$	Installed solar capacity	MWh
$\bar{K}^i$	Nameplate capacity of conventional producer $i \in \mathcal{N}^C$	MWh
a	Baseline offset in solar bid-quantity profile	MWh
b	Amplitude of solar bid-quantity profile	MWh
ρ	Penalty rate for wind under-delivery	\$/MWh
$mc^W \approx 0$	Marginal cost of wind generation	\$/MWh

8.3 Demand Parameters

Symbol	Description	Units
μ^D	Mean electricity demand, constant across all intervals	MWh
$(\sigma^D)^2$	Variance of electricity demand, constant across all intervals	MWh ²

8.4 Wind Production Parameters

Symbol	Description	Units
μ^W	Mean actual wind-power production	MWh
$(\sigma^W)^2$	Variance of actual wind-power production	MWh ²

9. Uncertain Parameters and Stochastic Variables

9.1 Stochastic Variables

These are quantities that are unknown at bid-submission time and realised only during or after market clearing.

Symbol	Description	Distribution	Scope
$\tilde{q}_t^W$	Actual wind-power production in interval t	$\mathcal{N}(\mu^W, (\sigma^W)^2)$	i.i.d. across $t \in \mathcal{T}$
$\tilde{D}_t$	Total market demand in interval t	$\mathcal{N}(\mu^D, (\sigma^D)^2)$	i.i.d. across $t \in \mathcal{T}$
$\tilde{p}_t^i$	Bid price of conventional producer i in interval t	$\mathcal{N}(\mu_p^i, (\sigma_p^i)^2)$	i.i.d. across $t \in \mathcal{T}$, for each $i \in \mathcal{N}^C$

Symbol	Description	Distribution	Scope
$\tilde{p}_t^S$	Bid price of solar producer in interval t	$\mathcal{N}(\mu_p^S, (\sigma_p^S)^2)$	i.i.d. across $t \in \mathcal{T}$
$\tilde{\lambda}_t$	Market-clearing price in interval t	Emergent from merit-order clearing	Determined each interval
$\tilde{e}_t^W$	Wind delivery shortfall in interval t	$\max(0, \tilde{q}_t^{W,\text{disp}} - \tilde{q}_t^W)$	Derived; realised post-dispatch

9.2 Uncertain Parameters

The means and variances of competitor bid-price distributions are not directly observable and must be estimated from historical market data. They are treated as fixed but unknown quantities:

Symbol	Description	Units
μ_p^i	Mean bid price of conventional producer $i \in \mathcal{N}^C$	\$/MWh
$(\sigma_p^i)^2$	Variance of bid price of conventional producer $i \in \mathcal{N}^C$	(\$/MWh) ²
μ_p^S	Mean bid price of solar producer	\$/MWh
$(\sigma_p^S)^2$	Variance of bid price of solar producer	(\$/MWh) ²

9.3 Support Constraints and Approximations

All normally distributed variables with a natural lower bound of zero are treated as valid approximations under the assumption that the coefficient of variation is small. Specifically:

- $\tilde{q}_t^W \geq 0$ requires $\sigma^W / \mu^W \ll 1$; otherwise a truncated normal on $[0, \bar{Q}^W]$ should be substituted
- $\tilde{D}_t \geq 0$ requires $\sigma^D / \mu^D \ll 1$; otherwise a truncated normal on $[0, \infty)$ should be substituted
- $\tilde{p}_t^i \geq 0$ requires $\sigma_p^i / \mu_p^i \ll 1$ for all i , consistent with cost-recovery bidding behaviour

9.4 Dependency Structure

All stochastic variables are mutually independent and i.i.d. across intervals:

$$\tilde{q}_t^W \perp \tilde{D}_t \perp \tilde{p}_t^i \perp \tilde{p}_t^S, \quad \forall i \in \mathcal{N}^C, \forall t \in \mathcal{T}$$

$$\tilde{q}_t^W \perp \tilde{q}_{t'}^W, \quad \forall t \neq t'$$

This is a deliberate simplification. In practice, wind output exhibits temporal autocorrelation, demand follows a diurnal cycle, and competitor prices may be correlated across intervals. These dependencies can be incorporated in subsequent extensions without altering the structural framework.

10. Modelling Assumptions

1. **No inter-temporal constraints:** ramp rates, start-up costs, minimum up/down times, and storage are excluded
2. **Perfect competition:** all producers are price-takers; no strategic manipulation of the supply stack
3. **Inelastic demand:** $\tilde{D}_t$ is exogenous and must be met exactly; no demand response or load shedding
4. **No transmission constraints:** the market is modelled as a single copper-plate node
5. **No balancing market:** imbalances are settled solely through the penalty mechanism at rate ρ ; no separate real-time market
6. **Bid commitment:** bids are submitted once and cannot be revised within the day

7. **Surplus wind not remunerated:** if $\tilde{q}_t^W > \tilde{q}_t^{W,\text{disp}}$, the excess is curtailed or spilled without payment

8. **Penalty applies to shortfall only:** the penalty is strictly one-directional over delivery (The wind-power producer should be remunerated at the clearing price for its full dispatched quantity, and then separately penalised for under-delivery. Please correct your specification accordingly.)

The correction affects Section 6.4 and the objective function in Section 7.1. Both are revised below.

6.4 Settlement and Penalty Assessment (Revised)

The wind producer is remunerated at the clearing price for its full dispatched quantity, irrespective of actual output, and then separately charged a penalty for any under-delivery:

$$r_t^W = \lambda_t \cdot \tilde{q}_t^{W,\text{disp}} - \rho \cdot \tilde{e}_t^W$$

where $\tilde{e}_t^W = \max(0, \tilde{q}_t^{W,\text{disp}} - \tilde{q}_t^W)$ is the shortfall defined in Section 6.3. The two terms are settled independently: the revenue term is determined entirely by the dispatch outcome and clearing price, while the penalty term is triggered only when physical delivery falls short of the dispatch obligation. The daily net revenue is:

$$R_d^W = \sum_{t=1}^{24} r_t^W = \sum_{t=1}^{24} \left(\lambda_t \cdot \tilde{q}_t^{W,\text{disp}} - \rho \cdot \max\left(0, \tilde{q}_t^{W,\text{disp}} - \tilde{q}_t^W\right) \right)$$

7.1 Wind Producer Objective (Revised)

The wind producer chooses its bid schedule to maximise expected daily net revenue:

$$\max_{\{q_t^W, p_t^W\}_{t \in \mathcal{T}}} \mathbb{E}[R_d^W] = \mathbb{E} \left[\sum_{t=1}^{24} \left(\lambda_t \cdot \tilde{q}_t^{W,\text{disp}} - \rho \cdot \max\left(0, \tilde{q}_t^{W,\text{disp}} - \tilde{q}_t^W\right) \right) \right]$$

subject to:

$$0 \leq q_t^W \leq \bar{Q}^W, \quad p_t^W \geq 0, \quad \forall t \in \mathcal{T}$$

The structure of the trade-off is unchanged: a higher bid quantity q_t^W increases the expected dispatched quantity and hence expected revenue, but also raises the expected shortfall and therefore expected penalty costs. However, under the corrected settlement rule, the revenue and penalty exposures are fully separable — the revenue component depends on $\tilde{q}_t^{W,\text{disp}}$ alone, while the penalty component depends on the gap $\tilde{q}_t^{W,\text{disp}} - \tilde{q}_t^W$. This separability is lost under the previous formulation, where revenue was contingent on actual output rather than dispatched quantity.

Various stakeholders approach this problem from different perspectives. Please revise your specification to use agents to represent two perspectives separately and independently: (1) the wind-power producer; and (2) the system regulator. In the revised specification, specify the common environment, including the agents, global constants, uncertain parameters and stochastic variables. For each perspective, specify the agent's key components, including state variables, control variables, processes and objectives.

Agent-Based Model Specification: Wind-Power Producer in a Day-Ahead Spot Market

Dual-Perspective Formulation

Part I: Common Environment

The common environment defines the shared market context within which both agent perspectives operate. It specifies the market structure, all agents, global constants, uncertain parameters, and stochastic variables. Neither perspective alters these shared elements; they form the exogenous backdrop against which each agent pursues its own objectives.

I.1 Time Scales

Scale	Definition
Planning horizon	A single calendar day d
Trading interval	Each hour $t \in \mathcal{T} = \{1, 2, \dots, 24\}$
Bid submission	Once per day, prior to interval $t = 1$, covering all $T = 24$ intervals simultaneously
Market clearing	Once per interval t , after all bids are received
Delivery and settlement	Once per interval t , after physical delivery; penalties assessed at settlement

I.2 Agents

Five producer agents and one market operator agent together constitute the environment. Two of these — the wind producer and the market operator — are additionally the subjects of the two perspectives defined in Part II and Part III respectively.

I.2.1 Producer Agents

Let $\mathcal{N} = \mathcal{N}^C \cup \{s\} \cup \{w\}$ denote the full set of five producers:

- $\mathcal{N}^C = \{c_1, c_2, c_3\}$ — conventional (coal) producers
- $\{s\}$ — solar producer
- $\{w\}$ — wind producer

Each producer submits a bid schedule $\{(q_t^i, p_t^i)\}_{t \in \mathcal{T}}$ once per day. Conventional and solar producers follow fixed exogenous behavioural rules. The wind producer's bidding strategy is the subject of Perspective 1.

I.2.2 Market Operator Agent

A passive, deterministic clearing agent that receives all bids, applies the merit-order mechanism with priority tie-breaking for each interval t , returns market-clearing prices and dispatch quantities, and enforces the penalty rule at settlement. The market operator's design — including the penalty rate and clearing rules — is the subject of Perspective 2.

I.3 Exogenous Bid Formation Rules

The following rules govern the two producer types whose behaviour is treated as exogenous to both perspectives.

Producer	Bid Quantity q_t^i	Bid Price p_t^i
Conventional $i \in \mathcal{N}^C$	$\bar{K}^i$ (fixed, full capacity)	$\tilde{p}_t^i \sim \mathcal{N}(\mu_p^i, (\sigma_p^i)^2)$, i.i.d. across t
Solar s	$\max(0, a + b \cos(2\pi t/24))$	$\tilde{p}_t^S \sim \mathcal{N}(\mu_p^S, (\sigma_p^S)^2)$, i.i.d. across t

I.4 Market Clearing Process

For each interval $t \in \mathcal{T}$, the market operator executes the following procedure.

Step 1 — Rank bids. Sort all five bids in ascending order of bid price. Ties are broken by the fixed priority ordering:

$$\pi : s \succ w \succ c_1 \succ c_2 \succ c_3$$

Let $\sigma(k)$ denote the producer at rank $k \in \{1, \dots, 5\}$ after sorting.

Step 2 — Identify the marginal unit. Define cumulative supply after the first k ranked producers as:

$$Q_k^{\text{cum}} = \sum_{j=1}^k q_t^{\sigma(j)}, \quad Q_0^{\text{cum}} \equiv 0$$

The marginal rank is:

$$k^* = \min \left\{ k : Q_k^{\text{cum}} \geq \tilde{D}_t \right\}$$

Step 3 — Set the clearing price:

$$\lambda_t = p_t^{\sigma(k^*)}$$

Step 4 — Determine dispatch quantities:

$$\tilde{q}_t^{\sigma(k), \text{disp}} = \begin{cases} q_t^{\sigma(k)} & \text{if } k < k^* \\ \tilde{D}_t - Q_{k^*-1}^{\text{cum}} & \text{if } k = k^* \\ 0 & \text{if } k > k^* \end{cases}$$

Step 5 — Set dispatch indicators:

$$\delta_t^i = \mathbf{1} \left[\tilde{q}_t^{i, \text{disp}} > 0 \right]$$

I.5 Settlement Process

The wind producer is remunerated at the clearing price for its full dispatched quantity and separately penalised for under-delivery. The shortfall and net revenue per interval are:

$$\tilde{e}_t^W = \max \left(0, \tilde{q}_t^{W, \text{disp}} - \tilde{q}_t^W \right)$$

$$r_t^W = \lambda_t \cdot \tilde{q}_t^{W, \text{disp}} - \rho \cdot \tilde{e}_t^W$$

Daily net revenue:

$$R_d^W = \sum_{t=1}^{24} r_t^W$$

I.6 Global Constants

I.6.1 Structural Constants

Symbol	Description	Units
$T = 24$	Number of trading intervals per day	—
$\$$	$\backslash\text{mathcal{N}}$	= 5\$
$\pi = (s, w, c_1, c_2, c_3)$	Fixed priority ordering for tie-breaking	—

I.6.2 Techno-Economic Parameters

Symbol	Description	Units
$\bar{Q}^W$	Installed wind capacity	MWh

Symbol	Description	Units
$\bar{Q}^S$	Installed solar capacity	MWh
$\bar{K}^i$	Nameplate capacity of conventional producer $i \in \mathcal{N}^C$	MWh
a	Baseline offset in solar bid-quantity profile	MWh
b	Amplitude of solar bid-quantity profile	MWh
$mc^W \approx 0$	Marginal cost of wind generation	\$/MWh

I.6.3 Demand Parameters

Symbol	Description	Units
μ^D	Mean electricity demand, constant across all intervals	MWh
$(\sigma^D)^2$	Variance of electricity demand, constant across all intervals	MWh ²

I.6.4 Wind Production Parameters

Symbol	Description	Units
μ^W	Mean actual wind-power production	MWh
$(\sigma^W)^2$	Variance of actual wind-power production	MWh ²

I.7 Uncertain Parameters

The means and variances of competitor bid-price distributions are not directly observable and must be estimated from historical market data. They are treated as fixed but unknown quantities, shared across both perspectives.

Symbol	Description	Units
μ_p^i	Mean bid price of conventional producer $i \in \mathcal{N}^C$	\$/MWh
$(\sigma_p^i)^2$	Variance of bid price of conventional producer $i \in \mathcal{N}^C$	(\$/MWh) ²
μ_p^S	Mean bid price of solar producer	\$/MWh
$(\sigma_p^S)^2$	Variance of bid price of solar producer	(\$/MWh) ²

I.8 Stochastic Variables

Symbol	Description	Distribution	Scope
$\tilde{q}_t^W$	Actual wind-power production in interval t	$\mathcal{N}(\mu^W, (\sigma^W)^2)$	i.i.d. across $t \in \mathcal{T}$
$\tilde{D}_t$	Total market demand in interval t	$\mathcal{N}(\mu^D, (\sigma^D)^2)$	i.i.d. across $t \in \mathcal{T}$
$\tilde{p}_t^i$	Bid price of conventional producer i in interval t	$\mathcal{N}(\mu_p^i, (\sigma_p^i)^2)$	i.i.d. across t , for each $i \in \mathcal{N}^C$
$\tilde{p}_t^S$	Bid price of solar producer in interval t	$\mathcal{N}(\mu_p^S, (\sigma_p^S)^2)$	i.i.d. across $t \in \mathcal{T}$
$\tilde{\lambda}_t$	Market-clearing price in interval t	Emergent from merit-order clearing	Determined each interval
$\tilde{e}_t^W$	Wind delivery shortfall in interval t	$\max(0, \tilde{q}_t^{W,\text{disp}} - \tilde{q}_t^W)$	Derived; realised post-dispatch

All stochastic variables are mutually independent and i.i.d. across intervals:

$$\tilde{q}_t^W \perp \tilde{D}_t \perp \tilde{p}_t^i \perp \tilde{p}_t^S \quad \forall i \in \mathcal{N}^C, \forall t \in \mathcal{T}$$

$$\tilde{q}_t^W \perp \tilde{q}_{t'}^W \quad \forall t \neq t'$$

Part II: Perspective 1 — Wind-Power Producer

This perspective models the wind producer as a strategic, optimising agent operating within the common environment. The producer takes all exogenous processes — competitor bids, demand, and market clearing — as given and chooses its own bid schedule to maximise expected net revenue.

II.1 State Variables

Variable	Symbol	Units	Description
Realised wind output	$\tilde{q}_t^W$	MWh	Actual generation available in interval t , revealed after bid submission
Dispatched quantity	$\tilde{q}_t^{W,\text{disp}}$	MWh	Quantity the producer is obligated to deliver, determined at clearing
Delivery shortfall	$\tilde{e}_t^W = \max(0, \tilde{q}_t^{W,\text{disp}} - \tilde{q}_t^W)$	MWh	Under-delivery relative to dispatch obligation in interval t
Dispatch indicator	$\delta_t^W \in \{0, 1\}$	—	1 if the wind producer is dispatched in interval t
Interval net revenue	r_t^W	\$	Revenue less penalty in interval t
Accumulated daily net revenue	$R_d^W = \sum_{\tau=1}^{t-1} r_\tau^W$	\$	Net revenue earned up to but not including interval t

II.2 Control Variables

The wind producer's bid schedule is its sole set of control variables, submitted once before $t = 1$:

$$\{(q_t^W, p_t^W)\}_{t \in \mathcal{T}}$$

Variable	Symbol	Units	Description	Constraint
Bid quantity	q_t^W	MWh	Quantity offered for supply in interval t	$0 \leq q_t^W \leq \bar{Q}^W$
Bid price	p_t^W	\$/MWh	Minimum acceptable price in interval t	$p_t^W \geq 0$

Given near-zero marginal cost, p_t^W is typically set at or near zero, making q_t^W the principal control variable. The commitment of q_t^W one day in advance, before $\tilde{q}_t^W$ is realised, is the central source of penalty risk.

II.3 Processes

The wind producer observes the following sequence of events each interval t :

1. **Bid commitment** (before $t = 1$): submit (q_t^W, p_t^W) for all $t \in \mathcal{T}$
2. **Market clearing** (at t): market operator determines λ_t , $\tilde{q}_t^{W,\text{disp}}$, and δ_t^W via the process defined in Section I.4
3. **Output realisation** (at t): $\tilde{q}_t^W \sim \mathcal{N}(\mu^W, (\sigma^W)^2)$ is revealed
4. **Shortfall calculation** (at t): $\tilde{e}_t^W = \max(0, \tilde{q}_t^{W,\text{disp}} - \tilde{q}_t^W)$
5. **Settlement** (at t): $r_t^W = \lambda_t \cdot \tilde{q}_t^{W,\text{disp}} - \rho \cdot \tilde{e}_t^W$

From the wind producer's perspective, the penalty rate ρ is an exogenous constant set by the regulator. The producer treats it as a given cost parameter when optimising its bid strategy.

II.4 Objective

The wind producer maximises expected daily net revenue over its bid schedule:

$$\max_{\{q_t^W, p_t^W\}_{t \in \mathcal{T}}} \mathbb{E}[R_d^W] = \mathbb{E} \left[\sum_{t=1}^{24} \left(\lambda_t \cdot \tilde{q}_t^{W, \text{disp}} - \rho \cdot \max \left(0, \tilde{q}_t^{W, \text{disp}} - \bar{q}_t^W \right) \right) \right]$$

subject to:

$$0 \leq q_t^W \leq \bar{Q}^W, \quad p_t^W \geq 0, \quad \forall t \in \mathcal{T}$$

A higher bid quantity q_t^W increases both the probability of dispatch and the expected dispatched quantity, raising expected revenue. It also increases the expected shortfall $\mathbb{E}[\tilde{e}_t^W]$, raising expected penalty costs. The optimal q_t^W balances these two opposing forces, with the penalty rate ρ governing the steepness of the penalty exposure.

Part III: Perspective 2 — System Regulator

This perspective models the market operator as a regulatory agent whose concern is the efficient and reliable operation of the electricity system as a whole. Rather than optimising a bid strategy, the regulator designs the penalty mechanism — specifically the penalty rate ρ — to align the wind producer's private incentives with socially desirable outcomes.

III.1 State Variables

Variable	Symbol	Units	Description
Market-clearing price	λ_t	\$/MWh	Clearing price in interval t , emergent from merit-order process
Aggregate dispatch schedule	$\{\tilde{q}_t^{i, \text{disp}}\}_{i \in \mathcal{N}}$	MWh	Dispatched quantities for all producers in interval t
Wind delivery shortfall	$\tilde{e}_t^W$	MWh	Under-delivery by the wind producer in interval t
System shortfall indicator	$\mathbf{1}[\tilde{e}_t^W > 0]$	—	1 if the wind producer fails to meet its dispatch obligation in interval t
Total daily wind shortfall	$E_d^W = \sum_{t=1}^{24} \tilde{e}_t^W$	MWh	Aggregate under-delivery across the day
Total daily penalty collected	$P_d^W = \rho \cdot E_d^W$	\$	Total penalty revenue collected from the wind producer

III.2 Control Variables

The regulator's sole control variable is the penalty rate:

Variable	Symbol	Units	Description	Constraint
Penalty rate	ρ	\$/MWh	Per-unit cost levied on the wind producer for each MWh of under-delivery	$\rho \geq 0$

The penalty rate is set once, prior to any market activity, and applies uniformly across all intervals $t \in \mathcal{T}$ and all future days. From the regulator's perspective, ρ is a policy instrument; from the wind producer's perspective (Part II), it is an exogenous constant.

III.3 Processes

The regulator observes the following sequence of events:

1. **Policy setting** (before market opens): set ρ , taking into account its anticipated effect on the wind producer's bidding behaviour $\{q_t^W(\rho)\}_{t \in \mathcal{T}}$, which is itself the solution to the wind producer's optimisation problem in Section II.4
2. **Market monitoring** (each interval t): observe λ_t , $\tilde{q}_t^{W, \text{disp}}$, $\bar{q}_t^W$, and $\tilde{e}_t^W$ following clearing and delivery
3. **Penalty enforcement** (each interval t): apply penalty $\rho \cdot \tilde{e}_t^W$ to the wind producer at settlement

4. **Outcome assessment** (end of day): aggregate shortfall E_d^W and penalty collected P_d^W are recorded for regulatory evaluation

The regulator accounts for the wind producer's rational response to ρ when setting policy, treating the bid quantity as a function $q_t^W(\rho)$ derived from the lower-level optimisation in Part II. This constitutes a Stackelberg or bilevel structure, in which the regulator is the leader and the wind producer is the follower.

III.4 Objective

The regulator minimises expected system costs arising from wind under-delivery, subject to maintaining sufficient incentive for the wind producer to participate in the market:

$$\min_{\rho \geq 0} \mathbb{E}[C_d(\rho)] = \mathbb{E} \left[\sum_{t=1}^{24} (c^{\text{short}} \cdot \tilde{e}_t^W(\rho)) \right]$$

where c^{short} (\$/MWh) is the social cost of undelivered electricity — representing, for instance, the value of lost load or the cost of emergency back-up generation — and $\tilde{e}_t^W(\rho) = \max(0, \tilde{q}_t^{W, \text{disp}}(\rho) - \tilde{q}_t^W)$ reflects the endogenous dependence of shortfall on ρ through the wind producer's bidding response.

The regulator's problem is subject to the wind producer's participation constraint:

$$\mathbb{E}[R_d^W(\rho)] \geq R^{\min}$$

where $R^{\min}$ is the minimum expected daily net revenue required for the wind producer to remain in the market. The penalty rate ρ thus faces a two-sided tension: setting ρ too low provides insufficient incentive for accurate bidding, increasing expected shortfall; setting ρ too high depresses the wind producer's expected net revenue, potentially violating the participation constraint or inducing overly conservative bids that forgo renewable generation.

An additional constant specific to this perspective is therefore introduced:

Symbol	Description	Units
c^{short}	Social cost of undelivered electricity (value of lost load)	\$/MWh
$R^{\min}$	Minimum expected daily net revenue for wind producer participation	\$

Your specification of the role of the system regulator is acceptable. However, you should not consider the system regulator and the market operator as the same agent, so there should be seven agents in this environment. Please refine your specification accordingly.

The correction requires revisions to Section I.2, which must now distinguish the market operator and system regulator as separate agents, and to the preamble of Part III, which previously conflated the two roles. All other sections remain unchanged.

I.2 Agents (*Revised*)

Seven agents populate the environment: five producer agents, one market operator agent, and one system regulator agent.

I.2.1 Producer Agents

Let $\mathcal{N} = \mathcal{N}^C \cup \{s\} \cup \{w\}$ denote the full set of five producers:

- $\mathcal{N}^C = \{c_1, c_2, c_3\}$ — conventional (coal) producers
- $\{s\}$ — solar producer
- $\{w\}$ — wind producer

Each producer submits a bid schedule $\{(q_t^i, p_t^i)\}_{t \in \mathcal{T}}$ once per day. Conventional and solar producers follow fixed exogenous behavioural rules. The wind producer's bidding strategy is the subject of Perspective 1.

I.2.2 Market Operator Agent

A passive, deterministic clearing agent that receives all bids, applies the merit-order mechanism with priority tie-breaking for each interval t , returns market-clearing prices and dispatch quantities, and administers the settlement process — including collection of penalty payments from the wind producer — in accordance with the rules established by the system regulator. The market operator exercises no discretion and pursues no objective of its own; it is purely an execution mechanism.

I.2.3 System Regulator Agent

A separate policy-setting agent responsible for designing the rules under which the market operator functions. In the current single-day formulation, the regulator's principal instrument is the penalty rate ρ , which it sets prior to any market activity. The regulator acts as a Stackelberg leader: it anticipates the rational bidding response of the wind producer when choosing ρ , and instructs the market operator to enforce the chosen rate at settlement. The system regulator's decision problem is the subject of Perspective 2.

Part III: Perspective 2 — System Regulator (*Revised Preamble*)

This perspective models the system regulator as a distinct policy-setting agent, separate from the market operator that operationally clears and settles the market. The regulator's role is not to run the market on a day-to-day basis — that function belongs to the market operator — but to design the penalty mechanism that governs it. Specifically, the regulator chooses the penalty rate ρ prior to market opening, with the objective of aligning the wind producer's private bidding incentives with socially desirable reliability outcomes. The market operator then enforces this rate mechanically at settlement, without further regulatory discretion.

The state variables, control variables, processes, and objective of the system regulator are otherwise unchanged from the previous specification.

To ensure internal consistency across these perspectives, please compose them into a unified agent-based model. The composed model should: 1. embed each perspective's components, including state variables, control variables, processes and objectives; and 2. enable modular evaluation of each perspective, where other perspectives' controls are treated as exogenous inputs. Please specify the key components of the final model according to the ODD protocol.

Agent-Based Model: Wind-Power Producer in a Day-Ahead Spot Market

Specification Following the ODD Protocol

SECTION 1: OVERVIEW

1.1 Purpose

The model simulates a single day of operations in a competitive day-ahead electricity spot market. Its primary purpose is to estimate the expected net revenue of a wind-power producer bidding under uncertainty about its own output and about competitor behaviour. A secondary purpose is to support the design of a penalty mechanism that aligns the wind producer's private bidding incentives with system-level reliability objectives.

The model is structured around two modular perspectives that can be evaluated independently:

- **Perspective 1 (Wind Producer):** given an exogenously fixed penalty rate ρ , optimise the wind producer's bid schedule to maximise expected daily net revenue

- Perspective 2 (System Regulator):** given the wind producer's rational bidding response, choose the penalty rate ρ to minimise expected system costs subject to the wind producer's participation constraint

When composed, the two perspectives form a bilevel optimisation structure in which the regulator acts as Stackelberg leader and the wind producer as follower. When evaluated in isolation, each perspective treats the other's control variable as an exogenous input, enabling modular analysis.

1.2 Entities, State Variables, and Scales

1.2.1 Scales

Scale	Definition
Planning horizon	A single calendar day d
Trading interval	Each hour $t \in \mathcal{T} = \{1, 2, \dots, 24\}$
Bid submission	Once per day, prior to $t = 1$, covering all $T = 24$ intervals
Market clearing	Once per interval t , after all bids are received
Delivery and settlement	Once per interval t , after physical delivery; penalties assessed at settlement

1.2.2 Agents and State Variables

Seven agents populate the model. Each agent's state variables are listed below.

Agent 1: Wind Producer w

The strategic, optimising agent whose bidding problem is the focus of Perspective 1.

Variable	Symbol	Units	Description
Bid quantity	q_t^W	MWh	Quantity offered for supply in interval t ; set before $t = 1$
Bid price	p_t^W	\$/MWh	Minimum acceptable price in interval t ; set before $t = 1$
Dispatched quantity	$\tilde{q}_t^{W,\text{disp}}$	MWh	Quantity the producer is obligated to deliver; determined at clearing
Dispatch indicator	$\delta_t^W \in \{0, 1\}$	—	1 if dispatched in interval t
Realised wind output	$\tilde{q}_t^W$	MWh	Actual generation available in interval t ; revealed after bid submission
Delivery shortfall	$\tilde{e}_t^W$	MWh	$\max(0, \tilde{q}_t^{W,\text{disp}} - \tilde{q}_t^W)$; under-delivery relative to obligation
Interval net revenue	r_t^W	\$	Revenue less penalty in interval t
Accumulated daily net revenue	R_d^W	\$	$\sum_{\tau=1}^{t-1} r_\tau^W$; net revenue earned up to but not including interval t

Agents 2-4: Conventional Producers c_1, c_2, c_3

Exogenous price-setting agents whose behaviour is fixed and independent of both perspectives.

For each $i \in \mathcal{N}^C$:

Variable	Symbol	Units	Description
Bid quantity	$q_t^i = \bar{K}^i$	MWh	Fixed at full installed capacity in every interval

Variable	Symbol	Units	Description
Bid price	$\tilde{p}_t^i$	\$/MWh	Drawn from $\mathcal{N}(\mu_p^i, (\sigma_p^i)^2)$, i.i.d. across t
Dispatched quantity	$\tilde{q}_t^{i,\text{disp}}$	MWh	Quantity dispatched; determined at clearing
Dispatch indicator	$\delta_t^i \in \{0, 1\}$	—	1 if dispatched in interval t

Agent 5: Solar Producer s

An exogenous producer with a deterministic quantity profile and stochastic bid price.

Variable	Symbol	Units	Description
Bid quantity	q_t^S	MWh	$\max(0, a + b \cos(2\pi t/24))$; deterministic function of t
Bid price	$\tilde{p}_t^S$	\$/MWh	Drawn from $\mathcal{N}(\mu_p^S, (\sigma_p^S)^2)$, i.i.d. across t
Dispatched quantity	$\tilde{q}_t^{S,\text{disp}}$	MWh	Quantity dispatched; determined at clearing
Dispatch indicator	$\delta_t^S \in \{0, 1\}$	—	1 if dispatched in interval t

Agent 6: Market Operator

A passive, deterministic execution agent. It holds no persistent state of its own beyond the interval-level outputs it produces and passes to other agents.

Variable	Symbol	Units	Description
Market-clearing price	λ_t	\$/MWh	Price resulting from merit-order clearing in interval t
Dispatch schedule	$\{\tilde{q}_t^{i,\text{disp}}\}_{i \in \mathcal{N}}$	MWh	Dispatched quantities for all producers in interval t
Marginal rank	k^*	—	Rank of the marginal producer in the sorted supply stack

Agent 7: System Regulator

The policy-setting agent whose design problem is the focus of Perspective 2.

Variable	Symbol	Units	Description
Penalty rate	ρ	\$/MWh	Per-unit penalty levied on wind under-delivery; set once before market opens
Observed wind shortfall	$\tilde{e}_t^W$	MWh	Under-delivery by the wind producer in interval t ; monitored post-settlement
Total daily wind shortfall	$E_d^W = \sum_{t=1}^{24} \tilde{e}_t^W$	MWh	Aggregate shortfall across the day
Total daily penalty collected	$P_d^W = \rho \cdot E_d^W$	\$	Total penalty revenue collected from the wind producer

1.2.3 Global Constants

Structural Constants

Symbol	Description	Units
$T = 24$	Number of trading intervals per day	—
\$	$\backslash\mathrm{mathcal{N}\{}$	= 5\$
$\pi = (s, w, c_1, c_2, c_3)$	Fixed priority ordering for tie-breaking	—

Techno-Economic Parameters

Symbol	Description	Units
$\bar{Q}^W$	Installed wind capacity	MWh
$\bar{Q}^S$	Installed solar capacity	MWh
$\bar{K}^i$	Nameplate capacity of conventional producer $i \in \mathcal{N}^C$	MWh
a	Baseline offset in solar bid-quantity profile	MWh
b	Amplitude of solar bid-quantity profile	MWh
$mc^W \approx 0$	Marginal cost of wind generation	\$/MWh

Demand Parameters

Symbol	Description	Units
μ^D	Mean electricity demand, constant across all intervals	MWh
$(\sigma^D)^2$	Variance of electricity demand	MWh ²

Wind Production Parameters

Symbol	Description	Units
μ^W	Mean actual wind-power production	MWh
$(\sigma^W)^2$	Variance of actual wind-power production	MWh ²

Regulator Parameters

Symbol	Description	Units
c^{short}	Social cost of undelivered electricity (value of lost load)	\$/MWh
$R^{\min}$	Minimum expected daily net revenue for wind producer participation	\$

1.2.4 Uncertain Parameters

Fixed but unobservable quantities estimated from historical data, shared across both perspectives.

Symbol	Description	Units
μ_p^i	Mean bid price of conventional producer $i \in \mathcal{N}^C$	\$/MWh
$(\sigma_p^i)^2$	Variance of bid price of conventional producer $i \in \mathcal{N}^C$	(\$/MWh) ²
μ_p^S	Mean bid price of solar producer	\$/MWh
$(\sigma_p^S)^2$	Variance of bid price of solar producer	(\$/MWh) ²

1.2.5 Stochastic Variables

Symbol	Description	Distribution	Scope
$\tilde{q}_t^W$	Actual wind-power production in interval t	$\mathcal{N}(\mu^W, (\sigma^W)^2)$	i.i.d. across $t \in \mathcal{T}$
$\tilde{D}_t$	Total market demand in interval t	$\mathcal{N}(\mu^D, (\sigma^D)^2)$	i.i.d. across $t \in \mathcal{T}$
$\tilde{p}_t^i$	Bid price of conventional producer i	$\mathcal{N}(\mu_p^i, (\sigma_p^i)^2)$	i.i.d. across t , for each $i \in \mathcal{N}^C$
$\tilde{p}_t^S$	Bid price of solar producer	$\mathcal{N}(\mu_p^S, (\sigma_p^S)^2)$	i.i.d. across $t \in \mathcal{T}$
$\tilde{\lambda}_t$	Market-clearing price in interval t	Emergent from merit-order clearing	Determined each interval

Symbol	Description	Distribution	Scope
$\tilde{e}_t^W$	Wind delivery shortfall in interval t	$\max(0, \tilde{q}_t^{W, \text{disp}} - \tilde{q}_t^W)$	Derived; realised post-dispatch

All primitive stochastic variables are mutually independent and i.i.d. across intervals:

$$\tilde{q}_t^W \perp \tilde{D}_t \perp \tilde{p}_t^i \perp \tilde{p}_t^S \quad \forall i \in \mathcal{N}^C, \forall t \in \mathcal{T}$$

$$\tilde{q}_t^W \perp \tilde{q}_{t'}^W \quad \forall t \neq t'$$

1.3 Process Overview and Scheduling

The model executes in three sequential phases per simulated day. Phases 1 and 2 execute once per day; Phase 3 executes once per interval and is repeated for all $t \in \mathcal{T}$.

```

PHASE 1 – PRE-MARKET POLICY SETTING [once, before t = 1]
|
| Regulator sets penalty rate ρ
| (Perspective 2 control; treated as exogenous input in Perspective 1)
|
└─ PHASE 2 – BID SUBMISSION [once, before t = 1]
    |
    | Conventional producers i ∈ N^C: q_t^i = K^i, p_t^i ~ N(μ_p^i,
    (σ_p^i)^2)
    | Solar producer s: q_t^S = max(0, a + b·cos(2πt/24)),
    p_t^S ~ N(μ_p^S, (σ_p^S)^2)
    | Wind producer w: submit (q_t^W, p_t^W) for all t ∈ T
    | (Perspective 1 controls; treated as exogenous inputs in Perspective 2)
    |
    └─ PHASE 3 – INTERVAL LOOP [repeat for each t = 1, ..., 24]
        |
        |─ Step 3.1 STOCHASTIC REALISATION
        |   D_t ~ N(μ^D, (σ^D)^2)
        |   q̃_t^W ~ N(μ^W, (σ^W)^2)
        |
        |─ Step 3.2 MARKET CLEARING [Market Operator]
        |   Rank bids by price, break ties via π = (s, w, c1, c2, c3)
        |   Find marginal rank k*: min{k : Q_k^cum ≥ D_t}
        |   Set clearing price: λ_t = p_{t}^{σ(k*)}
        |   Set dispatch quantities: q̃_t^{i,disp} for all i ∈ N
        |
        |─ Step 3.3 DELIVERY AND SHORTFALL [Wind Producer]
        |   Shortfall: e_t^W = max(0, q̃_t^{W,disp} - q̃_t^W)
        |
        |─ Step 3.4 SETTLEMENT [Market Operator, enforcing ρ]
        |   Interval net revenue: r_t^W = λ_t · q̃_t^{W,disp} - ρ ·
        e_t^W
        |
        |─ Step 3.5 STATE UPDATE
        |   R_d^W ← R_d^W + r_t^W
        |   E_d^W ← E_d^W + e_t^W
        |   P_d^W ← ρ · E_d^W
        |
        END OF DAY:
        |   Wind producer records R_d^W (Perspective 1 outcome)
        |   Regulator records E_d^W, P_d^W, and evaluates C_d(ρ) (Perspective 2
        outcome)

```

SECTION 2: DESIGN CONCEPTS

2.1 Basic Principles

The model is grounded in two foundational principles. The first is the **merit-order market clearing** mechanism, in which bids are stacked in ascending order of price and the cheapest combination of producers sufficient to meet demand is dispatched, with all accepted producers remunerated at the single clearing price of the marginal unit. This is the standard clearing principle of competitive day-ahead electricity markets. The second is **rational, self-interested optimisation**: the wind producer chooses its bid schedule to maximise its own expected payoff, and the regulator chooses the penalty rate to minimise social costs, each accounting for the other's rational response.

2.2 Emergence

The market-clearing price λ_t is an emergent quantity: it is not assigned to any agent but arises endogenously from the interaction of all five bid schedules and the realised demand level in each interval. The wind producer's revenue and the regulator's cost outcomes therefore cannot be determined without simulating the full clearing process. The distribution of λ_t across simulation runs is a key emergent output of the model.

2.3 Adaptation

The wind producer adapts its bid schedule in response to the penalty rate ρ set by the regulator. Specifically, the optimal bid quantity $q_t^W(\rho)$ is a function of ρ : as ρ increases, the producer faces greater expected penalty exposure and optimally reduces q_t^W toward its expected output μ^W . This adaptive response is the mechanism through which the regulator influences market outcomes, and it is the central behavioural linkage between the two perspectives.

2.4 Objectives

Each agent with a decision problem has a distinct objective:

- **Wind producer**: maximise expected daily net revenue $\mathbb{E}[R_d^W]$ over the bid schedule $\{(q_t^W, p_t^W)\}_{t \in \mathcal{T}}$
- **System regulator**: minimise expected daily system cost $\mathbb{E}[C_d(\rho)]$ over the penalty rate ρ , subject to the wind producer's participation constraint

The market operator and the conventional and solar producers have no objectives in the modelling sense; their behaviour is fully determined by fixed rules.

2.5 Learning

No learning mechanism is included in the single-day formulation. All agents act on the basis of fixed distributional assumptions about uncertain parameters. Learning across days — for instance, Bayesian updating of competitor bid-price distributions — is a natural extension for multi-day formulations.

2.6 Prediction

The wind producer implicitly predicts the market-clearing price λ_t when forming its bid. Under the price-taking assumption, it takes competitor bids and demand as drawn from fixed known distributions, and optimises over the resulting expected value of $\lambda_t \cdot \tilde{q}_t^{W, \text{disp}}$. The regulator similarly predicts the wind producer's optimal bid response $q_t^W(\rho)$ before setting ρ .

2.7 Sensing

Each agent observes only the information it requires for its own decision:

- **Wind producer**: knows its own bid, the clearing price λ_t , its dispatched quantity $\tilde{q}_t^{W, \text{disp}}$, and its realised output $\tilde{q}_t^W$; does not directly observe competitor bids
- **System regulator**: observes post-settlement outcomes $\tilde{e}_t^W$, E_d^W , and P_d^W ; knows the wind producer's optimisation problem and can predict $q_t^W(\rho)$
- **Market operator**: observes all submitted bids and the realised demand $\tilde{D}_t$; has no knowledge of post-delivery outcomes

2.8 Interaction

Agents interact indirectly through the market mechanism. No agent directly observes or responds to another agent's private state; all interactions are mediated by the market operator through the common clearing price and dispatch schedule. The regulator interacts

with the wind producer only through the policy instrument ρ , which enters the producer's payoff function as an exogenous parameter.

2.9 Stochasticity

Stochasticity enters the model through four independent channels: wind output $\tilde{q}_t^W$, market demand $\tilde{D}_t$, and the bid prices of conventional producers $\tilde{p}_t^i$ and the solar producer $\tilde{p}_t^S$. All are modelled as i.i.d. normal draws across intervals. The resulting stochasticity in λ_t and $\tilde{e}_t^W$ propagates into both agents' objective values.

2.10 Observation

The primary outputs observed from each simulation run are:

- **Perspective 1:** daily net revenue R_d^W ; its expectation and variance across runs; interval-level dispatch and revenue profiles
- **Perspective 2:** daily system cost $C_d(\rho) = c^{\text{short}} \cdot E_d^W$; total penalty collected P_d^W ; wind producer expected net revenue $\mathbb{E}[R_d^W(\rho)]$ as a function of ρ

2.11 Modularity

Each perspective is modular in the following sense. When evaluating Perspective 1, ρ is treated as a fixed scalar input and the regulator's optimisation is not executed. When evaluating Perspective 2, the wind producer's bid schedule $\{q_t^W(\rho)\}$ is treated as the solution to its lower-level problem and is computed as a function of ρ prior to simulation. The composed model activates both optimisations simultaneously, forming the bilevel structure described in Section 1.1.

SECTION 3: DETAILS

3.1 Initialisation

At the start of each simulated day, the following initialisation steps are performed.

Regulator initialisation (Perspective 2 active): the penalty rate ρ is set by solving the regulator's optimisation problem, anticipating the wind producer's best response. If Perspective 2 is inactive, ρ is set to an exogenously supplied scalar.

Wind producer initialisation (Perspective 1 active): the bid schedule $\{(q_t^W, p_t^W)\}_{t \in \mathcal{T}}$ is set by solving the wind producer's optimisation problem given ρ . If Perspective 1 is inactive, the bid schedule is set to an exogenously supplied vector.

Exogenous producer initialisation: for each $i \in \mathcal{N}^C$ and $t \in \mathcal{T}$, bid prices $\tilde{p}_t^i$ are drawn from $\mathcal{N}(\mu_p^i, (\sigma_p^i)^2)$. For the solar producer, $q_t^S = \max(0, a + b \cos(2\pi t/24))$ is computed deterministically and $\tilde{p}_t^S$ is drawn from $\mathcal{N}(\mu_p^S, (\sigma_p^S)^2)$.

Accumulator initialisation: $R_d^W \leftarrow 0, E_d^W \leftarrow 0, P_d^W \leftarrow 0$.

3.2 Input Data

The model requires no time-series input data in the single-day formulation. All stochastic processes are fully characterised by the distributional parameters listed in Sections 1.2.3 and 1.2.4, which are treated as calibrated constants. In future multi-day extensions, historical bid and price data would be used to estimate uncertain parameters $\{\mu_p^i, (\sigma_p^i)^2, \mu_p^S, (\sigma_p^S)^2\}$ and to validate the demand and wind production distributions.

3.3 Submodels

Submodel 1: Wind Producer Bid Optimisation (Perspective 1)

Given ρ and the distributional parameters of all stochastic variables, the wind producer solves:

$$\max_{\{q_t^W, p_t^W\}_{t \in \mathcal{T}}} \mathbb{E} \left[\sum_{t=1}^{24} \left(\lambda_t \cdot \tilde{q}_t^{W, \text{disp}} - \rho \cdot \max \left(0, \tilde{q}_t^{W, \text{disp}} - \tilde{q}_t^W \right) \right) \right]$$

$$\text{subject to } 0 \leq q_t^W \leq \bar{Q}^W, \quad p_t^W \geq 0, \quad \forall t \in \mathcal{T}$$

Since the objective is separable across intervals, this reduces to 24 independent single-interval problems. The expectation is taken over all stochastic variables, and $\tilde{q}_t^{W,\text{disp}}$ is itself a function of q_t^W, p_t^W , and the competitor bids and demand realisations. The solution $q_t^W(\rho)$ is the best-response function used by the regulator in Submodel 2.

Submodel 2: Regulator Penalty Design (Perspective 2)

Given the wind producer's best-response function $q_t^W(\rho)$, the regulator solves:

$$\min_{\rho \geq 0} \mathbb{E} \left[\sum_{t=1}^{24} c^{\text{short}} \cdot \max \left(0, \tilde{q}_t^{W,\text{disp}}(\rho) - \tilde{q}_t^W \right) \right]$$

$$\text{subject to } \mathbb{E}[R_d^W(\rho)] \geq R^{\min}$$

where $\tilde{q}_t^{W,\text{disp}}(\rho)$ is the dispatched quantity implied by the wind producer's optimal bid $q_t^W(\rho)$. The constraint ensures that the wind producer retains sufficient expected revenue to justify market participation. The composed bilevel problem is obtained by substituting Submodel 1 into Submodel 2 as the lower-level problem.

Submodel 3: Merit-Order Market Clearing

Executed by the market operator for each $t \in \mathcal{T}$:

$$k^* = \min \left\{ k : \sum_{j=1}^k q_t^{\sigma(j)} \geq \tilde{D}_t \right\}$$

$$\lambda_t = p_t^{\sigma(k^*)}$$

$$\tilde{q}_t^{\sigma(k),\text{disp}} = \begin{cases} q_t^{\sigma(k)} & k < k^* \\ \tilde{D}_t - Q_{k^*-1}^{\text{cum}} & k = k^* \\ 0 & k > k^* \end{cases}$$

Ties in bid price are resolved by the fixed priority ordering $\pi = (s, w, c_1, c_2, c_3)$.

Submodel 4: Settlement

Executed by the market operator after physical delivery in each interval t :

$$\tilde{e}_t^W = \max \left(0, \tilde{q}_t^{W,\text{disp}} - \tilde{q}_t^W \right)$$

$$r_t^W = \lambda_t \cdot \tilde{q}_t^{W,\text{disp}} - \rho \cdot \tilde{e}_t^W$$

Accumulators are updated:

$$R_d^W \leftarrow R_d^W + r_t^W, \quad E_d^W \leftarrow E_d^W + \tilde{e}_t^W, \quad P_d^W \leftarrow \rho \cdot E_d^W$$

3.4 Modelling Assumptions

1. **No inter-temporal constraints:** ramp rates, start-up costs, minimum up/down times, and storage are excluded
2. **Perfect competition:** all producers are price-takers; no strategic manipulation of the supply stack
3. **Inelastic demand:** $\tilde{D}_t$ is exogenous and must be met exactly; no demand response or load shedding
4. **No transmission constraints:** the market is modelled as a single copper-plate node

5. **No balancing market:** imbalances are settled solely through the penalty mechanism at rate ρ ; no separate real-time market
6. **Bid commitment:** bids are submitted once and cannot be revised within the day
7. **Surplus wind not remunerated:** if $\hat{q}_t^W > \hat{q}_t^{W, \text{disp}}$, the excess is curtailed without payment
8. **Penalty is strictly one-directional:** over-delivery carries no bonus beyond the revenue already earned on the dispatched quantity
9. **i.i.d. stochastic processes:** all stochastic variables are independent across intervals; temporal autocorrelation in wind output and demand is excluded at this stage

Now, please provide a modular Python implementation of the unified model, with each agent implemented as a separate class and a single interface function defined at the end.

All three perspectives run cleanly. Let me present the file.

The implementation runs correctly across all three perspectives. Here is a summary of the design decisions.

Structure. Seven classes map one-to-one onto the seven agents in the specification:

`WindProducer`, `ConventionalProducer` ($\times 3$), `SolarProducer`, `MarketOperator`, and `SystemRegulator`. Shared data flows between agents via three lightweight dataclasses — `Bid`, `ClearingResult`, and `SettlementResult` — rather than through shared global state, preserving agent encapsulation.

Submodel mapping. Each numbered submodel from the ODD specification has a direct method counterpart:

`WindProducer.optimise_bids` (Submodel 1), `SystemRegulator.optimise_penalty` (Submodel 2), `MarketOperator.clear_interval` (Submodel 3), and `MarketOperator.settle_interval` (Submodel 4).

Optimisation. Submodel 1 exploits the separability of the wind producer's objective across intervals: each of the 24 bid quantities is optimised independently using `scipy.optimize.minimize_scalar` with bounded domain $[0, Q^W]$, with the expectation estimated by Monte Carlo. Submodel 2 uses a coarse grid search followed by bounded scalar refinement, calling Submodel 1 at each candidate p to embed the wind producer's best response.

Modularity. The `perspective` argument to `run_simulation` controls which optimisations are active. In `perspective1`, p is treated as an exogenous scalar. In `perspective2`, wind bids are fixed at μ^W . In `composed`, the full bilevel structure is solved. In all cases the simulation loop (`_run_single_day`) and the agent classes are identical — only the initialisation phase differs.



Wind market abm

PY

Download

Please correct the following two parts in your implementation: (1) Solar bid profile should use 1-indexed instead of 0-indexed; and (2) Clearing price not updated when the marginal unit is fully dispatched. In `clear_interval`, `clearing_price` is only updated in the else branch (partial dispatch).

Two surgical fixes are required. Both are confined to single methods.

Fix 1 — `SolarProducer.bid_quantity`: the cosine argument uses `t` (0-indexed) but the specification defines $t \in \{1, \dots, 24\}$, so `t` should be replaced with `t + 1`.

Fix 2 — `MarketOperator.clear_interval`: `clearing_price` and `marginal_producer` are only reassigned inside the `else` branch (partial dispatch). When the marginal unit is fully dispatched — i.e. its quantity exactly meets the remaining demand — the loop ends via `break` after the `if` branch without updating either variable, leaving them at their

initialised values (the last element of the stack). Both must be updated in the `if` branch as well.

The file has been partially edited and now has an inconsistency: `bid_quantity` already adds `+1` internally, but `get_bid` and `sample_competitor_bids` also pass `t+1`, creating a double offset. I'll apply all three corrections cleanly.

Now verify the clearing-price fix is consistent and check line 670:

Good — the clearing-price fix is already correctly applied. Now do a final smoke-test:

All assertions pass. Here is a precise account of every change made.

Changes made

Fix 1 — Solar bid profile (`SolarProducer`)

Three locations were involved:

- `bid_quantity(t)`: the method now computes `cos(2π(t+1)/24)` internally, converting the 0-indexed simulation interval to the 1-indexed specification interval. This is the single source of truth for the profile.
- `get_bid(t)`: corrected from `self.bid_quantity(t + 1)` to `self.bid_quantity(t)`. The previous version applied the offset twice, producing a phase shift of two hours at every call.
- `sample_competitor_bids(t, ...)`: corrected from `self.solar_producer.bid_quantity(t + 1)` to `self.solar_producer.bid_quantity(t)` for the same reason.

Fix 2 — Clearing price for fully-dispatched marginal unit (`MarketOperator.clear_interval`)

The loop previously only updated `clearing_price` and `marginal_producer` in the `else` branch (partial dispatch). When a unit's quantity exactly matched the residual demand — i.e. `bid.quantity <= residual` and `cumulative` reached `demand` after the update — neither variable was reassigned, leaving them at their initialised fallback values (the last element of the sorted stack). The fix adds a conditional update inside the `if` branch:

```
python

if bid.quantity <= residual:
    dispatched[pid] = bid.quantity
    cumulative += bid.quantity
    if cumulative >= demand:          # ← new: exact-match case
        marginal_producer = pid
        clearing_price = bid.price
```

This covers all three dispatch outcomes correctly: infra-marginal (fully dispatched, demand not yet met), exact-marginal (fully dispatched, demand exactly met), and partial-marginal (partially dispatched).

 Wind market abm
PY

Download

    < 2/2 >

